BRAC University
Dept. of Computer Science and Engineering
Spring 2025
CSE331 - Assignment 2
Deadline: May 16 11:45 PM

1. **Use the pumping lemma to demonstrate that the following languages are not regular.**
    a. L = { w ∈ $\{0,1\}^*$: ww^R and w^R means w in reverse}
    b. L = { w ∈ $\{a,b\}^*$: w = $b^n a^m$ where n > m ,m ≥ 0}
    c. L = { w ∈ $\{0,1,2,3\}^*$: w = $1^n 0^m 3^n 2^m$ where n,m ≥ 0}
2. **Give context-free grammars that generate the languages.**
    a. L = { x#y : x,y ∈ $\{0,1\}^*$ and |x| = |y| and the length of x is odd}
    b. L = { w ∈ $\{a,b\}^*$: w contains odd numbers of b's}
    c. L = { w ∈ $\{a,b,c\}^*$: w = $b^n a^m c^p$ where n > 2m + 4 , m,p ≥ 0}
3. **Design pushdown automata for the following languages.**
    a. L = { w ∈ $\{0,1\}^*$: w = $0^n 1^m$ where n > 3m ,m ≥ 0}
    b. L = { w ∈ $\{a,b\}^*$: w = $b^n a^m c^p$ where m = 2n, p is even ,m ≥ 0}
    c. L = { w ∈ $\{0,1\}^*$: w starts and ends with different symbols}